

Highlights

VitaTouch: A Property-Aware Vision–Tactile–Language Foundation Model for Manufacturing Quality Inspection

Junyi Zong, Qingxuan Jia, Meixian Shi, Tong Li, Jiayuan Li, Zihang Lv, Gang Chen, Fang Deng

- VitaSet: 186 objects, 52k images across modalities, 5.1k QA pairs
- Three-stage training: alignment, property learning, and LoRA adaptation
- VitaTouch: state-of-the-art on TVL; 92/73/52% hardness/roughness/material
- Few-shot LoRA: 80% defect accuracy with 15 samples per object
- Robotic sorting validates actionable defect decisions in lab setting

VitaTouch: A Property-Aware Vision–Tactile–Language Foundation Model for Manufacturing Quality Inspection

Junyi Zong^{a,e}, Qingxuan Jia^a, Meixian Shi^a, Tong Li^{a,*}, Jiayuan Li^{b,e},
Zihang Lv^a, Gang Chen^a, Fang Deng^{c,d,**}

^a*School of Intelligent Engineering and Automation, Beijing University of Posts and
Telecommunications, Beijing, 100876, China*

^b*School of Automation, Beijing Institute of Technology, Beijing, 100081, China*

^c*School of AI, Beijing Institute of Technology, Beijing, 100081, China*

^d*State Key Lab of Autonomous Intelligent Unmanned Systems, Beijing Institute of
Technology, Beijing, 100081, China*

^e*Zhongguancun Academy, Beijing, 100094, China*

Abstract

Quality inspection in smart manufacturing requires recognizing intrinsic material and surface properties beyond visible geometry, yet vision-only approaches struggle to infer these properties under imaging nuisances. To address these limitations, we propose **VitaTouch**, a **property-aware vision–tactile–language (VTL) foundation model** that integrates visual and tactile sensing with language prompts in a unified semantic space, enabling property inference and natural-language attribute descriptions. VitaTouch employs modality-specific encoders and dual Q-Formers to distill language-relevant evidence from vision and touch into compact prefix tokens, which are projected into the embedding space of a large language model (LLM) to condition generation. To strengthen cross-modal grounding, we align each modality with text and explicitly couple vision and touch via a contrastive objective, promoting unified multi-property reasoning. We build **VitaSet**, a VTL dataset with 186 objects, 52k images across modalities, and 5.1k instruction–answer pairs with human-verified labels. Extensive experi-

*Corresponding author.

**Co-corresponding author.

Email addresses: tli@bupt.edu.cn (Tong Li), dengfang@bit.edu.cn (Fang Deng)

ments demonstrate that VitaTouch achieves state-of-the-art performance on the TVL benchmark and improves property recognition on VitaSet (92.06% hardness, 73.28% roughness, and 52.06% material). With parameter-efficient LoRA adaptation, VitaTouch transfers to few-shot defect detection, reaching 80% accuracy with 15 labeled samples per object, and enables closed-loop robotic sorting in a lab quality-inspection workflow. Code and dataset are available on <https://vitatouch.github.io/>.

Keywords: Smart manufacturing, Defect detection, Large language model, Vision–tactile sensing, Multimodal foundation model, Few-shot learning

1. Introduction

Quality inspection is a cornerstone of smart manufacturing, underpinning product reliability, production efficiency, and lifecycle cost. In industrial practice, quality inspection is not a single-stage decision; it requires multi-level understanding of manufactured objects, including material property recognition during incoming inspection, monitoring of process-relevant attributes during production, and final defect inspection before delivery. Collectively, these tasks determine whether a product meets functional specifications, whether the process remains within tolerance, and whether early-stage degradations can be detected before they evolve into critical failures. From a systems perspective, inspection strategy is tightly coupled with quality-related cost and operational performance, motivating increasing interest in data-driven inspection policies under Industry 4.0 settings [1].

Despite rapid progress, mainstream industrial inspection pipelines remain predominantly vision-based. Modern machine-vision systems and deep models have achieved strong performance on a range of industrial benchmarks, especially under controlled conditions and with abundant training data. Large-scale anomaly detection datasets and surveys further consolidate the effectiveness of deep industrial anomaly detection (IAD) pipelines for image-level and pixel-level inspection [2, 3]. Meanwhile, systematic reviews of surface defect inspection using deep object detection networks highlight the maturity of vision-based solutions and their widespread adoption [4]. However, when quality inspection is viewed as multi-level physical understanding rather than appearance-based pattern recognition, purely visual pipelines exhibit intrinsic bottlenecks.

Vision alone provides an incomplete and fragile basis for physical quality assessment. It primarily captures geometry and photometric appearance, whereas many manufacturing-relevant cues are physical properties that are challenging to access in vision-only inspections, such as **hardness**, **roughness**, and **material**. Moreover, deployment in real shop-floor environments further amplifies failure modes: **occlusion** and visually inaccessible regions, **weak/uneven illumination**, and **specular reflections** on glossy or metallic surfaces can obscure defects and confound appearance cues. These issues are repeatedly observed as key pain points for deploying purely visual inspection in practice [4, 2]. Thus, improving reliability requires stepping beyond appearance-only cues toward sensing modalities that can directly capture the physical essence of manufactured surfaces.

Advances in tactile sensing and its growing adoption provide a new paradigm for addressing the above limitations by introducing complementary physical evidence. Vision-based tactile sensors (VBTS) capture contact-induced deformation patterns that encode surface and material property cues while producing RGB tactile readouts. Compared with conventional vision, these signals provide richer local contact details than conventional vision and reduce sensitivity to illumination artifacts that often hinder visual imaging [10]. Recent progress in visuo-tactile representation learning further suggests the feasibility of learning transferable tactile foundations across sensors and data regimes [11]. These developments motivate property-centric inspection that is grounded in contact mechanics, enabling more direct access to the physical attributes that drive quality assessment and early defect discovery.

In parallel, foundation models and large language models (LLMs) have reshaped multimodal learning by enabling a unified semantic interface between low-level sensory inputs and high-level concepts. Vision-language pretraining (e.g., CLIP) establishes scalable alignment between images and language, supporting open-vocabulary recognition and flexible prompting [5]. More recent multimodal LLM frameworks (e.g., BLIP-2, InstructBLIP, LLaVA) further enable instruction-following and open-ended generation conditioned on sensory evidence [6, 7, 8]. For inspection, this paradigm shift is particularly attractive because it moves beyond closed-set classification toward language-driven workflows, where operators can specify criteria in natural language and models can respond with open-form attribute descriptions and reasoning-based explanations when visual evidence is ambiguous. Meanwhile, early efforts have begun to incorporate touch into language-grounded perception, including touch-vision-language alignment via the TVL dataset and

models [30], as well as tactile–language models for object property reasoning, such as Octopi [31]. However, these works are primarily developed for general robotic perception and everyday objects rather than industrial quality inspection, and they do not explicitly target the stringent requirements of manufacturing deployment. Consistent with recent benchmarks, current multimodal LLMs still fall short of industrial requirements for anomaly- and defect-related understanding [9]. More fundamentally, property-centric physical understanding grounded in tactile evidence has not yet been established as a reusable foundation for smart-manufacturing quality inspection.

Practical deployment further demands that such foundations be adaptable and deployable under industrial constraints. In many production lines, labeled defect data are limited and fragmented across product variants, and collecting representative defect samples is difficult due to low incidence rates and long-tail variability. Furthermore, industrial inspection systems often face strict requirements for inference latency and reliability, being deployed on industrial PCs or edge devices means limited GPU memory and compute budgets. These constraints make full-model finetuning impractical, especially when incorporating large multimodal models. Therefore, effective solutions must support few-shot adaptation with minimal labeled data, and must do so via resource-aware mechanisms that enable real-time inference and robust deployment on constrained hardware. Existing methods, however, typically lack property-aware multimodal grounding, which can render few-shot transfer brittle when appearance cues shift or when defects are primarily characterized by physical-property changes rather than visual patterns.

This paper argues that **multimodal physical-property understanding** is a missing foundation for reliable quality inspection. We propose **VitaTouch**, a **vision–tactile–language (VTL) foundation model** designed to learn shared grounding among visual observations, tactile evidence, and natural-language queries, enabling multi-property inference and language-form outputs for material and surface understanding. VitaTouch follows a property-centric pretraining to deployment-oriented adaptation paradigm: it first establishes cross-modal grounding across vision, touch, and language; then learns reusable representations of material and surface properties; and finally supports efficient downstream adaptation to defect inspection under few-shot and resource-constrained settings via parameter-efficient updates (e.g., low-rank adaptation). By treating defect inspection as a downstream task built upon a general, property-aware foundation, VitaTouch better aligns with the multi-level nature of industrial quality inspection and

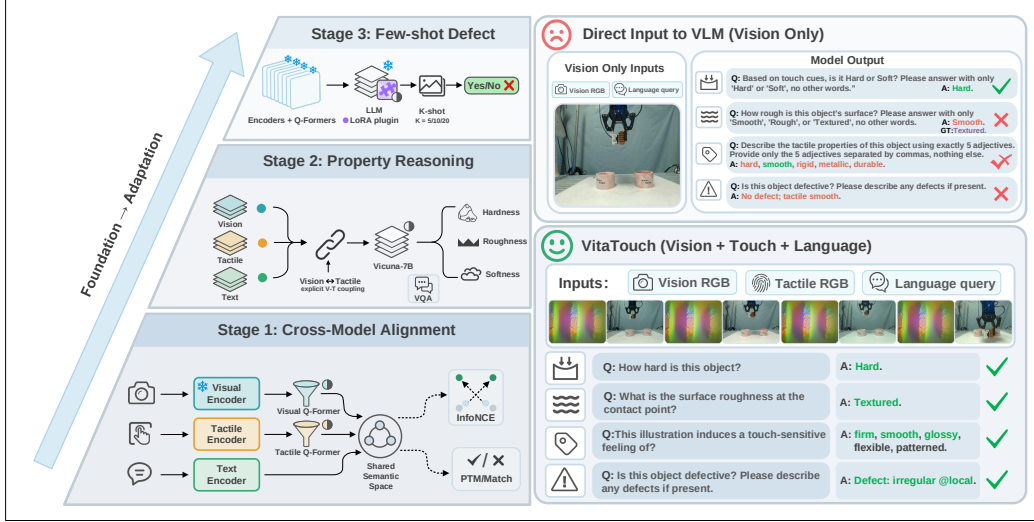


Figure 1: **Overview of VitaTouch.** Left: three-stage training pipeline. Stage 1: cross-modal alignment via dual Q-Formers (InfoNCE + PTM). Stage 2: property-reasoning foundation model with fused V-T tokens in frozen Vicuna-7B. Stage 3: few-shot defect adaptation using LoRA with K samples per object. Right: tactile sensing enables accurate property reasoning and defect detection where vision-only VLMs fail.

enables effective transfer from property-centric understanding to practical defect detection.

Contributions. Our main contributions are summarized as follows:

- We collect and release a high-quality vision-tactile-language (VTL) dataset in a robotic manipulation setting, with aligned visual observations, tactile signals, and language annotations to support training and evaluation for multimodal quality inspection.
- We propose a property-centric training pipeline that couples cross-modal alignment with multi-property attribute learning in the VTL setting, establishing a reusable backbone for material and surface property understanding.
- We introduce VitaTouch, a property-aware vision-tactile-language foundation model for multimodal quality inspection in smart manufacturing, targeting unified physical understanding rather than appearance-only detection.

- We demonstrate efficient transfer to few-shot defect detection under resource constraints via parameter-efficient adaptation, validating that property-centric pretraining supports practical industrial inspection scenarios.

The rest of the article is organized as follows. Section 3 introduces VitaSet, the training and evaluation suite for industrial quality inspection. Section 4 presents the VitaTouch architecture and the progressive three-stage training protocol. Section 5 demonstrates experimental results on public benchmarks and VitaSet, together with ablation study, few-shot defect adaptation, and real-world sorting demonstrations. Finally, we conclude the paper with a discussion of limitations and future directions.

2. Related Work

2.1. Vision-Based Defect Inspection with Deep Learning

Vision-based inspection has long been the dominant paradigm in industrial quality inspection, and deep learning has substantially improved the accuracy and scalability of defect detection and localization. When dense labels are available, supervised detection and segmentation models provide strong performance in controlled settings. However, in many production lines, defective samples are rare, diverse, and costly to annotate, which has fueled the rapid development of industrial anomaly detection (IAD) approaches that relies primarily on normal data. Representative IAD lines include feature distribution modeling (e.g., PaDiM) [12], discriminative reconstruction and embedding learning (e.g., DRAEM) [13], and retrieval-based patch embedding matching (e.g., PatchCore) [14]. Large-scale benchmarks such as MVTec AD [3] have become standard testbeds, while subsequent datasets aim to improve realism and coverage. Examples include VisA, introduced with the SPD pretraining framework [15], and Real-IAD for real-world multi-view inspection [16]. Comprehensive surveys and systematic reviews further summarize the maturity of deep IAD and surface defect inspection pipelines [2, 4]. Despite strong progress, purely visual pipelines remain sensitive to shop-floor factors such as occlusion, illumination changes, and specular reflections, and they are fundamentally limited when defects correlate more with deviations in physical property than stable appearance cues.

2.2. From Closed-Set Recognition to Language-Driven Inspection with LLM/VLMs

The recent surge of large language models (LLMs) and vision-language models (VLMs) is reshaping inspection, shifting from closed-set classification and detection to language-driven problem solving. Compared with conventional multi-class classifiers, VLM- and LLM-based systems offer (i) instruction following via natural-language queries, (ii) open-form descriptions and attribute reporting beyond a fixed label taxonomy, and (iii) semantic reasoning that can support interactive, operator-centric workflows. CLIP-style vision-language pretraining establishes scalable image-text alignment and enables promptable open-vocabulary recognition [5]. Building on this, multi-modal LLM frameworks (e.g., BLIP-2, InstructBLIP, LLaVA) connect visual evidence to strong language decoders, enabling instruction tuning and generative responses conditioned on multimodal observations [6, 7, 8]. However, manufacturing inspection has domain-specific requirements (e.g., fine-grained localization, long-tail categories, safety-critical precision), and recent benchmarks suggest that even state-of-the-art multimodal LLMs still fall short for anomaly and defect-centric understanding under industrial conditions [9]. These observations motivate targeted research on leveraging language capabilities while respecting industrial constraints.

2.3. VLM/LLM for Defect and Anomaly Detection: Methods and Benchmarks

A fast-growing literature investigates how to adapt CLIP-style VLMs for anomaly and defect detection and segmentation, highlighting language as a scalable mechanism for generalization across product categories. A key challenge is that generic VLMs emphasize object semantics rather than normality or abnormality semantics, making naive prompting insufficient for anomaly localization. WinCLIP addresses this issue with window- and patch-level feature aggregation and compositional prompt ensembling, enabling strong zero-/few-shot anomaly classification and segmentation on MVTec AD and VisA [17]. Prompt learning further reduces reliance on manual prompt engineering. For example, PromptAD proposes one-class prompt learning using only normal samples, improving few-shot anomaly detection under realistic supervision [18]. AnomalyCLIP learns object-agnostic prompts to capture generic normal/abnormal concepts and enhances cross-domain zero-shot anomaly detection [19]. Beyond prompt learning, context-conditioned prompting and cross-modal interactions have been explored to improve generalization. Researches such as VCP-CLIP, which injects visual context into

prompts to avoid product-specific prompt design [20], and AdaCLIP, a zero-shot anomaly detection (ZSAD) method that combines static and dynamic learnable prompts [21].

Recent work also composes multiple foundation models to improve localization quality. ClipSAM combines CLIP-style anomaly localization with SAM-based mask refinement to obtain sharper anomaly boundaries [22]. Meanwhile, anomaly-aware alignment has been studied to improve the discriminability of CLIP features for normal versus abnormal semantics: AA-CLIP proposes anomaly-aware text anchors and patch-level alignment while maintaining generalization [23], and FE-CLIP injects frequency cues into CLIP for improved zero-shot anomaly detection and segmentation [24]. Complementary to CLIP-centric pipelines, LLM-style industrial anomaly systems have emerged that emphasize dialogue and threshold-free outputs; for example, AnomalyGPT leverages large vision–language models with anomaly-oriented prompting and synthetic supervision to support multi-turn reporting and localization [25]. Collectively, methods evaluated on benchmarks such as MVTec AD [3], VisA [15], Real-IAD [16], and industrial MLLM benchmark MMAD [9], demonstrate that language can substantially enhance scalability, flexibility, and interpretability in defect and anomaly inspection.

2.4. Introducing Touch: Deep Visuo-Tactile Learning and LLM-Grounded Touch–Vision–Language

While language-guided VLMs improve semantic flexibility, reliable quality inspection also requires sensing modalities that capture physical surface and material cues beyond appearance. Vision-based tactile sensors (VBTS) provide image-like tactile observations that encode micro-geometry and contact mechanics and are increasingly feasible in practical systems [10]. Representative hardware platforms include GelSight [27] and the compact DIGIT sensor [28]. On the learning side, tactile representation learning aims to build transferable encoders across sensors and tasks. For instance, AnyTouch, studies unified static-dynamic representations across multiple visuo-tactile sensors [11]. In industrial contexts, multimodal inspection pipelines that combine global vision with local tactile scanning have shown practical value for detecting small defects on large-scale components, where a coarse-to-fine strategy mitigates coverage challenges and visual blind spots [29].

Importantly, touch research is now evolving from deep visuo-tactile fusion toward LLM-grounded tactile reasoning with natural language. The TVL dataset is a milestone, providing large-scale vision–touch pairs with language

annotations and pseudo-labels, which enables alignment of tactile encoders with vision–language spaces and tactile-conditioned text generation [30]. Oc-topi further demonstrates tactile-language property reasoning by combining tactile representation learning with VLMs and exploiting intermediate property predictions (e.g., hardness, roughness) to improve downstream reasoning [31]. Nevertheless, these milestones are primarily developed for general robotics and everyday object understanding, and industrial inspection still lacks a reusable, property-centric touch–vision–language foundation designed around manufacturing defect taxonomies, process variability, and deployment constraints.

2.5. Few-Shot Adaptation and Parameter-Efficient Tuning for Industrial Deployment

Deployment constraints for industrial quality inspection differ from general-purpose multimodal reasoning in several respects. **First**, defect data are scarce: failure modes are rare, diverse across product lines, and expensive to label due to domain-expertise requirements. **Second**, production environments are dynamic: new variants and process drift emerge continuously, requiring rapid model updates without exhaustive retraining. **Third**, computational budgets are tight: on-site inference ran on edge devices (e.g., industrial PCs) under strict latency and memory constraints, making deployment of large multimodal models challenging.

To address these constraints, different learning paradigms have been proposed. For data and task constraints, few-shot learning and meta-learning aim for rapid adaptation to new tasks using limited examples; representative methods include MAML [32] and prototypical networks [33]. For resource constraints, parameter-efficient tuning provides a practical path to adapting foundation models under compute and memory limits by updating only small trainable modules while keeping backbones frozen. LoRA, for example, injects trainable low-rank parameters into pretrained models and significantly reduces adaptation cost [34]. Most few-shot anomaly inspection approaches remain vision-centric and thus lack property-aware multimodal grounding. When defects correspond to physical property deviations rather than stable visual patterns, models without tactile evidence and explicit property understanding can become brittle under distribution shifts, motivating multimodal property-centric pretraining to support robust few-shot transfer in manufacturing.

2.6. Positioning

To summarize, prior work has (i) established strong vision-based IAD methods and benchmarks [3, 14, 13, 12, 15, 16, 2, 4]; (ii) advanced language-guided anomaly/defect detection by adapting CLIP/VLMs via prompting, prompt learning, context injection, and foundation-model composition [17, 18, 19, 20, 21, 22, 23, 24]; and (iii) introduced touch sensing, tactile representation learning, and emerging touch–vision–language milestones that enable tactile-grounded language interaction and reasoning [10, 27, 28, 11, 29, 30, 31]. These trends motivate building a property-aware vision–tactile–language foundation for industrial inspection that unifies open-language interaction, physically grounded perception, and resource-aware adaptation to few-shot defect tasks.

3. Dataset and Training & Evaluation Suite

This section introduces **VitaSet**, a vision–tactile–language (VTL) dataset and training/evaluation suite designed for industrial quality inspection. Existing visuo-tactile resources are mostly collected around everyday objects and general robotic perception, and they provide limited coverage of industrial operating conditions and inspection-oriented supervision. To bridge this gap, we collect an industrial-scene robotic manipulation dataset using a GelSight-equipped manipulator and further expand coverage by integrating the GelSight-only subset of AnyTouch. Under a unified annotation schema, VitaSet supports property-centric pretraining and inspection-oriented evaluation for our VitaTouch model.

3.1. Physical Property Selection

Industrial quality inspection verifies whether products conform to physical specifications that determine functionality and process stability. We therefore focus on three inspection-relevant properties that are (i) well-defined and measurable, (ii) frequently implicated in industrial nonconformities, and (iii) naturally supported by contact sensing in addition to RGB appearance. **Hardness.** Hardness quantifies resistance to indentation/deformation under applied pressure. In manufacturing workflows, hardness is a standard incoming-inspection attribute and an in-process conformity indicator; deviations can signal material mismatch or process drift even when RGB appearance remains inconclusive.

Roughness. Roughness characterizes surface micro-geometry that governs friction, wear, sealing, and coating adhesion. In shop-floor conditions, purely visual roughness cues can be unstable due to illumination variation, specular reflections, and low-texture surfaces. GelSight contact images provide direct evidence of micro-geometry changes under contact, complementing RGB observations.

Material category. Material category constrains expected mechanical and tribological behavior and is central to inspection procedures such as incoming checks, mixed-material prevention, and production-line sorting. VitaSet operationalizes material-related supervision in an inspection-reporting style: the model is asked to output a fixed-length list of tactile descriptors (e.g., top-5 adjectives from a controlled vocabulary) that best match the perceived material/surface feel, which enables auditable and actionable reporting for quality-inspection decisions.

3.2. Dataset Construction and Annotation

We release a dataset

$$\mathcal{D} = \{(\mathbf{X}_i, T_i, Y_i)\}_{i=1}^N, \quad \mathbf{X}_i = (X_i^v, X_i^t), \quad (1)$$

where X_i^v is the RGB image, X_i^t is the GelSight tactile image (when available), T_i is the instruction/query, and Y_i is the corresponding answer. We cast both discrete labels and descriptive outputs into text-form answers, enabling unified instruction-following supervision across tasks.

VitaSet integrates two complementary sources. First, we collect an industrial-scene robotic manipulation dataset using a GelSight-equipped manipulator under inspection-like interactions and operating conditions. Second, we incorporate the GelSight-only portion of AnyTouch (AnyTouch contains multiple sensors; we retain only its GelSight subset to ensure modality consistency). This integration increases diversity in object instances and contact outcomes while keeping the tactile modality aligned for model training.

In total, VitaSet contains **186 objects**, with **30,553** RGB images and **21,510** GelSight tactile images, yielding **52,063** images across modalities. We provide **5,145** annotated instruction-answer (QA) pairs. The tactile image count is smaller primarily because we manually removed a substantial number of tactile frames that do not correspond to valid object contact (e.g., no-contact or low-quality contact), whereas the RGB stream remains available for the corresponding trials.

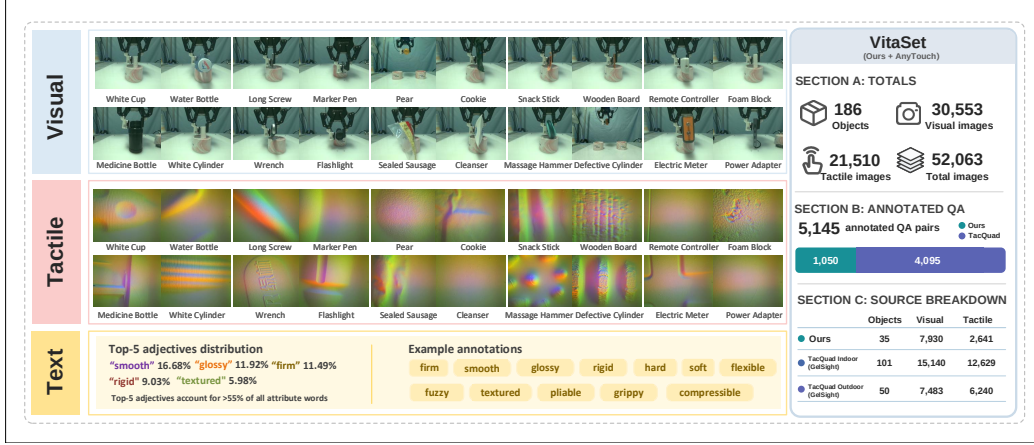


Figure 2: **VitaSet overview (Ours + AnyTouch GelSight-only)**. Aligned RGB observations and paired GelSight tactile readings across objects, with controlled-vocabulary annotations and dataset statistics under a unified schema.

We adopt a scalable yet auditable annotation pipeline. We first generate initial instruction-answer pairs using the **GPT-4o** API with template-constrained outputs: classification tasks are restricted to a predefined label set, and description tasks are restricted to a controlled vocabulary with a fixed-length format. We then perform complete object-level human verification over all **186 objects** (not partial sampling), reviewing and correcting generated labels to remove inconsistencies, implausible descriptions, and format violations. This full-coverage verification yields an auditable, unified label space shared by both the robotic collection and the AnyTouch-derived samples.

3.3. Training and Evaluation Suite

We define a suite of tasks aligned with quality-inspection workflows. All tasks are cast into an instruction-following form $(\mathbf{X}, T) \rightarrow Y$.

Task 1: Hardness classification (binary). Given (X^v, X^t) and a hardness query, the model outputs one token from {Hard, Soft}. **Metric:** Accuracy and F1 (reported when class imbalance exists).

Task 2: Roughness classification (3-way). Given (X^v, X^t) and a roughness query, the model outputs one token from {Smooth, Textured, Rough}. **Metric:** Accuracy (optionally complemented with class-wise accuracy / confusion analysis for adjacent-category ambiguity).

Task 3: Property description with controlled vocabulary. Given (X^v, X^t) and an instruction requesting tactile/property attributes, the model outputs a fixed-length list of adjectives selected only from the controlled vocabulary (e.g., top-5 descriptors). **Metric:** exact set matching, reported as precision/recall/F1 between the predicted adjective set and the ground-truth set (exact match only; no free-form text similarity scoring).

Task 4: Defect decision and brief defect description. Given (X^v, X^t) and a defect-related instruction, the model outputs a binary decision token (yes/no) and an optional short description when a defect is present. **Metric:** decision Accuracy/F1/AUROC parsed from the decision token; descriptions are evaluated qualitatively with inspection-style examples.

Overall, this suite couples property-centric physical understanding (Tasks 1–3) with inspection-oriented defect reporting (Task 4), supporting foundation training and downstream adaptation under manufacturing constraints.

4. Method

VitaTouch is a vision–tactile–language model for property reasoning and few-shot defect inspection. As shown in Fig. 1, we train VitaTouch with a progressive three-stage protocol: (i) cross-modal alignment to establish a shared semantic interface across vision, touch, and language; (ii) property-reasoning foundation learning by conditioning a frozen LLM on fused vision–tactile tokens; and (iii) parameter-efficient few-shot adaptation to defect detection using LoRA. The overall training procedure is summarized in the pseudo-code at the end of this section.

4.1. Problem Formulation

Each sample consists of an RGB image, a tactile image, and a natural-language query,

$$\mathcal{X}_i = (X_i^v, X_i^t, T_i), \quad (2)$$

where $X_i^v \in \mathbb{R}^{H_v \times W_v \times 3}$ denotes the visual observation, $X_i^t \in \mathbb{R}^{H_t \times W_t \times 3}$ denotes the tactile observation, and T_i specifies either a property query or a defect query. Given $\mathcal{X}_i$, VitaTouch generates a textual response Y_i .

4.2. Model Architecture

An overview of the proposed architecture is provided in Fig. 3. VitaTouch follows a dual-branch design with modality-specific encoders and Q-Formers.

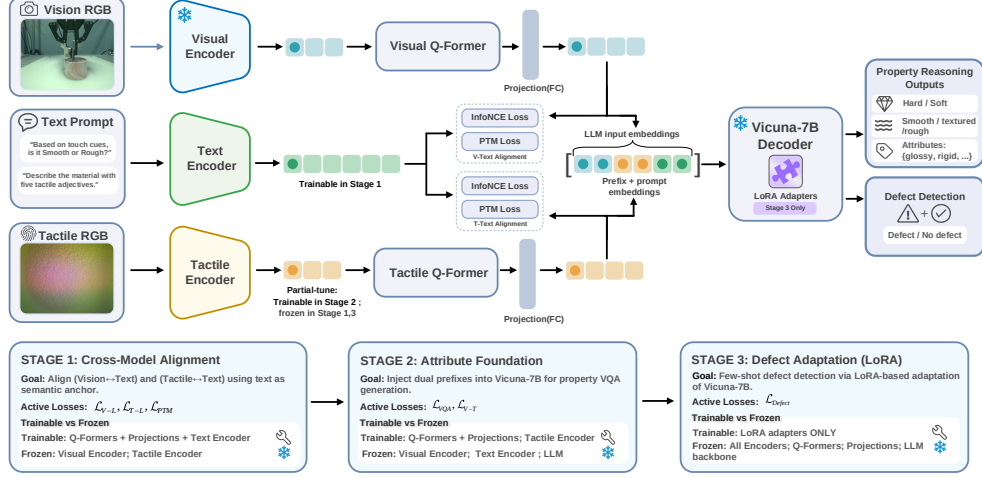


Figure 3: **VitaTouch model architecture.** VitaTouch employs a **dual-branch vision-tactile design** with modality-specific encoders and Q-Formers. The Q-Formers distill learnable queries into **vision-tactile prefix tokens**, which are prepended to text embeddings and fed into a frozen Vicuna-7B decoder. Training proceeds in three stages: **Stage 1** aligns cross-modal embeddings via frozen encoders; **Stage 2** establishes the perception-to-language pathway for property reasoning; **Stage 3** adapts the model for few-shot defect detection using LoRA while freezing the backbone.

The Q-Formers distill compact, language-relevant tokens from vision and touch, which are then mapped into the LLM embedding space as prefix tokens.

4.2.1. Perception Encoders

Given X_i^v , the visual encoder produces a token sequence

$$\mathbf{E}_i^v = f_{\theta}^v(X_i^v) \in \mathbb{R}^{L_v \times d_v}. \quad (3)$$

Given X_i^t , the tactile encoder produces

$$\mathbf{E}_i^t = f_{\theta}^t(X_i^t) \in \mathbb{R}^{L_t \times d_t}. \quad (4)$$

The visual encoder is frozen throughout all stages. The tactile encoder is frozen in Stage 1 and fully fine-tuned in Stage 2, while it is kept frozen in Stage 3.

4.2.2. Dual Q-Former

For each modality $m \in \{v, t\}$, we instantiate an independent Q-Former with L_q learnable query vectors,

$$\mathbf{Q}_0^m \in \mathbb{R}^{L_q \times d_q}, \quad L_q = 32. \quad (5)$$

Given perception tokens $\mathbf{E}_i^m$, the Q-Former outputs

$$\mathbf{Q}_i^m = \text{QF}^m(\mathbf{Q}_0^m, \mathbf{E}_i^m) \in \mathbb{R}^{L_q \times d_q}. \quad (6)$$

Each query token is projected into a shared embedding space and ℓ_2 -normalized:

$$\hat{\mathbf{q}}_{ik}^m = \frac{\mathbf{W}_p^m \mathbf{Q}_{ik}^m}{\|\mathbf{W}_p^m \mathbf{Q}_{ik}^m\|_2}, \quad \hat{\mathbf{q}}_{ik}^m \in \mathbb{R}^d, \quad k = 1, \dots, L_q. \quad (7)$$

4.2.3. Text Encoding and LLM Interface

Given query text T_i , a text encoder produces token states $\mathbf{Z}_i$. We project the [CLS] state into the shared embedding space:

$$\hat{\mathbf{t}}_i = \frac{\mathbf{W}_t \mathbf{Z}_{i, [\text{CLS}]}}{\|\mathbf{W}_t \mathbf{Z}_{i, [\text{CLS}]}\|_2} \in \mathbb{R}^d. \quad (8)$$

For Stage 2/3, query tokens are mapped into the LLM embedding space:

$$\tilde{\mathbf{Q}}_i^v, \tilde{\mathbf{Q}}_i^t \in \mathbb{R}^{L_q \times d_{\text{LLM}}}, \quad \tilde{\mathbf{Q}}_i^{vt} = [\tilde{\mathbf{Q}}_i^v; \tilde{\mathbf{Q}}_i^t] \in \mathbb{R}^{2L_q \times d_{\text{LLM}}}. \quad (9)$$

4.3. Stage 1: Cross-Modal Alignment Pretraining

Stage 1 learns a shared semantic interface by aligning each modality with language. For a minibatch of size N , we pool the query set by averaging:

$$\hat{\mathbf{z}}_i^m = \frac{1}{L_q} \sum_{k=1}^{L_q} \hat{\mathbf{q}}_{ik}^m \in \mathbb{R}^d, \quad m \in \{v, t\}. \quad (10)$$

We apply symmetric bidirectional InfoNCE between modality m and language:

$$\mathcal{L}_{m-L} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp(\langle \hat{\mathbf{z}}_i^m, \hat{\mathbf{t}}_i \rangle / \tau)}{\sum_{j=1}^N \exp(\langle \hat{\mathbf{z}}_i^m, \hat{\mathbf{t}}_j \rangle / \tau)} + \log \frac{\exp(\langle \hat{\mathbf{t}}_i, \hat{\mathbf{z}}_i^m \rangle / \tau)}{\sum_{j=1}^N \exp(\langle \hat{\mathbf{t}}_i, \hat{\mathbf{z}}_j^m \rangle / \tau)} \right] \quad (11)$$

The overall contrastive loss is defined as the sum of the modality–language contrastive terms:

$$\mathcal{L}_{\text{con}} = \mathcal{L}_{\text{v-L}} + \mathcal{L}_{\text{t-L}}. \quad (12)$$

To encourage semantic compatibility beyond metric alignment, we use perceptual-text matching with hard negatives:

$$\mathcal{L}_{\text{match}} = -\frac{1}{M} \sum_{\ell=1}^M \left[y_{\ell} \log p_{\ell} + (1 - y_{\ell}) \log(1 - p_{\ell}) \right], \quad (13)$$

The Stage 1 objective is

$$\mathcal{L}_{\text{Stage1}} = \mathcal{L}_{\text{con}} + \lambda_{\text{match}} \mathcal{L}_{\text{match}}. \quad (14)$$

4.4. Stage 2: Property-Reasoning Foundation Model

Stage 2 trains the model to answer property-related queries conditioned on fused vision–tactile prefixes. Given a property prompt T_i^{prop} and target sequence Y_i , we minimize the autoregressive loss:

$$\mathcal{L}_{\text{VQA}} = -\frac{1}{N} \sum_{i=1}^N \sum_{s=1}^{L_i} \log p_{\theta}(y_{i,s} \mid y_{i,<s}, X_i^{\text{v}}, X_i^{\text{t}}, T_i^{\text{prop}}). \quad (15)$$

We further enforce explicit vision–tactile coupling with a symmetric InfoNCE loss computed between the pooled vision and tactile embeddings:

$$\mathcal{L}_{\text{v-t}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp(\langle \hat{\mathbf{z}}_i^{\text{v}}, \hat{\mathbf{z}}_i^{\text{t}} \rangle / \tau_{\text{vt}})}{\sum_{j=1}^N \exp(\langle \hat{\mathbf{z}}_i^{\text{v}}, \hat{\mathbf{z}}_j^{\text{t}} \rangle / \tau_{\text{vt}})} + \log \frac{\exp(\langle \hat{\mathbf{z}}_i^{\text{t}}, \hat{\mathbf{z}}_i^{\text{v}} \rangle / \tau_{\text{vt}})}{\sum_{j=1}^N \exp(\langle \hat{\mathbf{z}}_i^{\text{t}}, \hat{\mathbf{z}}_j^{\text{v}} \rangle / \tau_{\text{vt}})} \right], \quad (16)$$

where τ_{vt} is a temperature hyperparameter.

The Stage 2 objective is

$$\mathcal{L}_{\text{Stage2}} = \mathcal{L}_{\text{VQA}} + \lambda_{\text{VT}} \mathcal{L}_{\text{v-t}}. \quad (17)$$

4.5. Stage 3: Few-Shot Defect Detection via LoRA Adaptation

Stage 3 adapts the model to defect detection. Given a defect prompt T_i^{def} , we optimize:

$$\mathcal{L}_{\text{defect}} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log p(\text{yes} \mid \mathcal{X}_i) + (1 - y_i) \log p(\text{no} \mid \mathcal{X}_i) \right]. \quad (18)$$

We apply LoRA to Vicuna attention projections:

$$\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r} \mathbf{B} \mathbf{A}. \quad (19)$$

At inference, we compute the decision score:

$$\Delta_i = \text{logit}(\text{yes} \mid \mathcal{X}_i) - \text{logit}(\text{no} \mid \mathcal{X}_i). \quad (20)$$

Overall training procedure. For completeness, Algorithm 1 summarizes our three-stage training pipeline. Stage 1 learns language-anchored cross-modal alignment between vision, touch, and text (InfoNCE + PTM). Stage 2 trains the property-reasoning foundation model by conditioning a frozen LLM on fused vision–tactile prefix tokens and optimizing an instruction-following objective with an additional vision–tactile coupling loss. Stage 3 performs parameter-efficient adaptation to defect detection by inserting LoRA modules into the LLM and updating only the LoRA parameters.

5. Experiments

Our experiments follow a progressive validation pipeline from public benchmarking to manufacturing-oriented evaluation and deployment. We first evaluate VitaTouch on the public TVL benchmark to assess its overall capability and enable direct comparison with prior methods. We then report multi-task property understanding on our self-collected VitaSet dataset tailored for manufacturing-relevant attributes, followed by an ablation study to analyze the contribution of key components. Next, we evaluate parameter-efficient few-shot defect detection via LoRA adaptation under limited labels. We further provide qualitative inspection-style outputs to illustrate format compliance and interpretability. Finally, we demonstrate a closed-loop robotic sorting system to verify that model predictions are actionable in a physical quality-inspection workflow.

5.1. Datasets and Benchmarks

TVL benchmark. We evaluate on the Touch–Vision–Language (TVL) benchmark [30], which provides a standardized protocol for comparing vision–tactile–language models. This benchmark provides a standardized protocol for evaluating vision–tactile–language models and enables direct comparison with prior approaches under a common evaluation setup.

Table 1: VitaTouch training procedure (three-stage pipeline)

1	Require: training set $\mathcal{D} = \{(X_i^v, X_i^t, T_i)\}$; property prompts T_i^{prop} with targets Y_i ; defect prompts T_i^{def} with labels y_i .
2	Initialize learnable queries $\mathbf{Q}_0^v, \mathbf{Q}_0^t$ and projection layers; freeze visual encoder f_θ^v throughout training and freeze Vicuna-7B.

3	Stage 1: Cross-modal alignment (InfoNCE + PTM).
4	for each minibatch $\{(X_i^v, X_i^t, T_i)\}_{i=1}^N$ do (freeze tactile encoder f_θ^t)
5	$\mathbf{E}_i^v \leftarrow f_\theta^v(X_i^v)$; $\mathbf{E}_i^t \leftarrow f_\theta^t(X_i^t)$.
6	$\mathbf{Q}_i^m \leftarrow \text{QF}^m(\mathbf{Q}_0^m, \mathbf{E}_i^m)$; $\hat{\mathbf{q}}_{ik}^m \leftarrow \ell_2\text{-norm}(\mathbf{W}_p^m \mathbf{Q}_{ik}^m)$, $m \in \{v, t\}$.
7	$\mathbf{Z}_i \leftarrow \text{Enc}^{\text{text}}(T_i)$; $\mathbf{t}_i \leftarrow \ell_2\text{-norm}(\mathbf{W}_t \mathbf{Z}_i)$.
8	Compute pooled embeddings $\hat{\mathbf{z}}_i^m \leftarrow \frac{1}{L_q} \sum_{k=1}^{L_q} \hat{\mathbf{q}}_{ik}^m$ (Eq. (10)); set $\mathcal{L}_{\text{con}} \leftarrow \mathcal{L}_{v-L} + \mathcal{L}_{t-L}$ (Eq. (12)).
9	$\mathcal{L}_{\text{match}} \leftarrow \text{Eq. (13)}$; $\mathcal{L}_{\text{Stage1}} \leftarrow \mathcal{L}_{\text{con}} + \lambda_{\text{match}} \mathcal{L}_{\text{match}}$.
10	Update Stage 1 trainable parameters by $\nabla \mathcal{L}_{\text{Stage1}}$.
11	end for

12	Stage 2: Property reasoning foundation (VQA + V-T coupling).
13	Unfreeze and fine-tune tactile encoder f_θ^t in Stage 2; keep f_θ^v and Vicuna-7B frozen.
14	for each minibatch $\{(X_i^v, X_i^t, T_i^{\text{prop}}, Y_i)\}_{i=1}^N$ do
15	Compute prefixes $\mathbf{Q}_i^{\text{vt}} = [\mathbf{Q}_i^v; \tilde{\mathbf{Q}}_i^t]$ and prepend to T_i^{prop} .
16	$\mathcal{L}_{\text{VQA}} \leftarrow \text{Eq. (15)}$; $\mathcal{L}_{v-t} \leftarrow \text{vision-tactile InfoNCE (Eq. (16))}$.
17	$\mathcal{L}_{\text{Stage2}} \leftarrow \mathcal{L}_{\text{VQA}} + \lambda_{\text{VT}} \mathcal{L}_{v-t}$.
18	Update Stage 2 trainable parameters by $\nabla \mathcal{L}_{\text{Stage2}}$ (including f_θ^t , Q-Formers, and projections).
19	end for

20	Stage 3: Few-shot defect adaptation (LoRA).
21	Insert LoRA into Vicuna attention projections; freeze f_θ^v , f_θ^t , Q-Formers, and projection layers.
22	for each minibatch $\{(X_i^v, X_i^t, T_i^{\text{def}}, y_i)\}_{i=1}^N$ do
23	Decode distribution over $\{\text{yes}, \text{no}\}$ conditioned on $(\tilde{\mathbf{Q}}_i^{\text{vt}}, T_i^{\text{def}})$.
24	$\mathcal{L}_{\text{defect}} \leftarrow \text{Eq. (18)}$; update LoRA parameters only.
25	end for

Self-collected VitaSet dataset. To better reflect smart manufacturing inspection requirements, we collect a synchronized vision–tactile–language dataset with an RGB camera and a GelSight tactile sensor. The dataset targets manufacturing-relevant physical understanding and supports the following tasks: (i) hardness perception (binary), (ii) roughness recognition (3-way), and (iii) material property description (tactile attribute generation). We also support language-form outputs for operator-facing reporting (controlled-vocabulary attributes / short descriptions), consistent with the Stage 2 instruction-following formulation.

5.2. Implementation Details

VitaTouch is implemented in PyTorch. We use EVA-CLIP-G as the visual encoder and an AnyTouch encoder (ViT-L backbone) for tactile inputs. The language backbone is Vicuna-7B.

Stage 1: cross-modal alignment. Stage 1 performs explicit cross-modal alignment. We train only the two modality-specific Q-Formers (vision and tactile) together with their projection layers that map query tokens into a shared embedding space, while keeping the perception encoders fixed. We use AdamW with a linear-warmup cosine learning-rate schedule, using a peak learning rate of 1×10^{-4} , warmup learning rate of 1×10^{-6} , 500 warmup steps, and weight decay of 0.01. Automatic mixed precision (AMP) is enabled throughout training.

Stage 2: property-reasoning foundation learning. Stage 2 performs multi-task instruction supervision for property understanding. In this stage, we freeze the LLM (Vicuna-7B) and the visual encoder, and finetune the tactile encoder, both Q-Formers, and the modality-to-LLM projection layers end-to-end to build a property-aware foundation model. We use AdamW with the same linear-warmup cosine schedule (peak LR 1×10^{-4} , warmup LR 1×10^{-6} , 500 warmup steps, weight decay 0.01) and AMP for stable large-model training. We set the maximum input/output length to 512 tokens to support inspection-style instruction outputs.

Stage 3: few-shot defect adaptation. Stage 3 adapts the Stage 2 foundation model to defect detection under limited labels. We freeze the perception encoders, Q-Formers, and projection layers, and apply LoRA to the Vicuna attention projections, updating only the LoRA parameters. We adopt a LoRA configuration with rank $r = 16$, scaling factor $\alpha = 32$, and dropout rate of 0.1 for efficient adaptation.

Training hardware and time. Stage 1 and Stage 2 are trained on an 8×NVIDIA A800 GPU cluster, taking approximately 2 hours (Stage 1) and 10 hours (Stage 2), respectively. Stage 3 is lightweight and can be completed within 1 hour on a single RTX 5090 GPU, which aligns with practical industrial deployment constraints.

5.3. Main Results

5.3.1. Results on the TVL Benchmark

We first evaluate VitaTouch on the public TVL benchmark to assess its capability under a standardized evaluation protocol and enable direct comparison with prior methods. Table 2 shows that VitaTouch achieves the best scores on **HCT** and **TVL** among the compared methods, while remaining

Table 2: Comparison on the TVL benchmark. VitaTouch achieves the best performance on HCT and TVL, and remains competitive on SSVTP.

Model	Encoder Pre-training Modalities			Score (1–10)		
	Vision	Tactile	Language	SSVTP	HCT	TVL
LLaVA-1.5 7B	✓	-	✓	3.64	3.55	3.56
LLaVA-1.5 13B	✓	-	✓	3.55	3.63	3.62
ViP-LLaVA 7B	✓	-	✓	2.72	3.44	3.36
ViP-LLaVA 13B	✓	-	✓	4.10	3.76	3.80
LLaMA-Adapter	✓	-	✓	2.56	3.08	3.02
BLIP-2 Opt-6.7b	✓	-	✓	2.02	2.72	2.64
InstructBLIP 7B	✓	-	✓	1.40	1.30	1.31
InstructBLIP 13B	✓	-	✓	1.44	1.21	1.24
GPT-4V	✓	-	✓	5.02	4.42	4.49
SSVTP-LLaMA	✓	✓	-	2.58	3.67	3.54
TVL-LLaMA (ViT-Tiny)	✓	✓	✓	6.09	4.79	4.94
TVL-LLaMA (ViT-Small)	✓	✓	✓	5.81	4.77	4.89
TVL-LLaMA (ViT-Base)	✓	✓	✓	6.16	4.89	5.03
VitaTouch (Ours)	✓	✓	✓	5.83	6.10	6.09

competitive on **SSVTP**. These results indicate that our dual Q-Former design and tri-modal grounding provide more effective vision-tactile-language alignment and property-centric reasoning under the TVL evaluation protocol.

5.3.2. Property Recognition on the VitaSet Dataset

To evaluate VitaTouch’s capability in industrial quality inspection scenarios, we train the model on our self-collected VitaSet dataset targeting manufacturing-relevant physical property understanding. Figure 4 reports the validation performance across training epochs for all tasks in our multi-task property learning setting. The curves demonstrate stable multi-task optimization and consistent convergence across hardness classification, roughness recognition, and material property description, indicating that a single VitaTouch foundation model can effectively support multiple property-

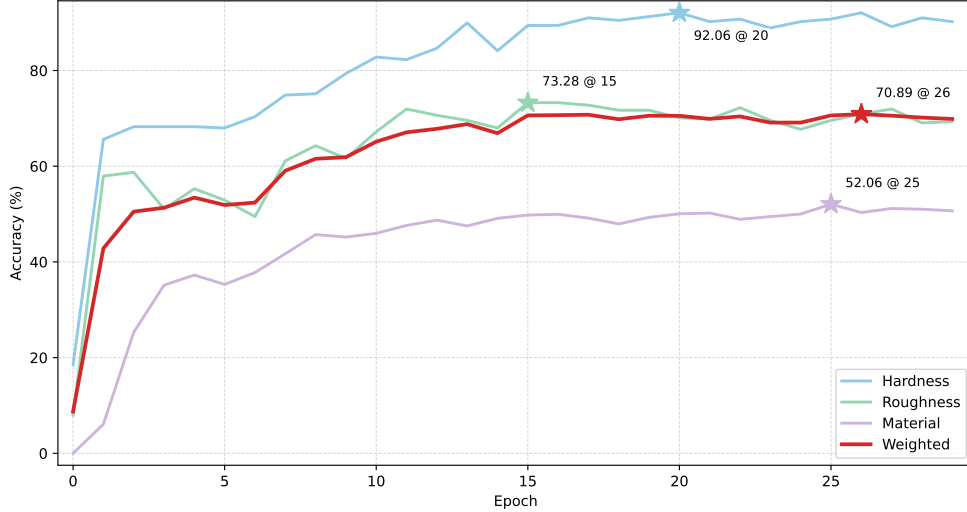


Figure 4: VitaSet dataset: validation performance vs. epoch across tasks. Stars denote the best epoch for each task/aggregate metric.

Table 3: Ablation results on the VitaSet dataset comparing unimodal baselines with VitaTouch.

Method	Hardness	Roughness	Material	Average
Vision only (Qwen-2.5-VL)	73.81%	61.11%	32.28%	55.73%
Tactile only (Octopi)	68.25%	58.20%	8.40%	44.95%
VitaTouch (Without Stage1 alignment)	89.42%	61.11%	6.67%	51.33%
VitaTouch (Ours)	92.06%	73.28%	52.06%	70.89%

reasoning tasks simultaneously. As shown in the figure, VitaTouch achieves strong performance across all tasks, demonstrating the effectiveness of our multi-modal fusion approach for manufacturing-relevant property understanding.

5.3.3. Ablation Study

To isolate the contribution of each component in VitaTouch, we conduct comprehensive ablation studies comparing the full model against unimodal baselines and variants without Stage 1 alignment. Table 3 and Figure 5 present the results.

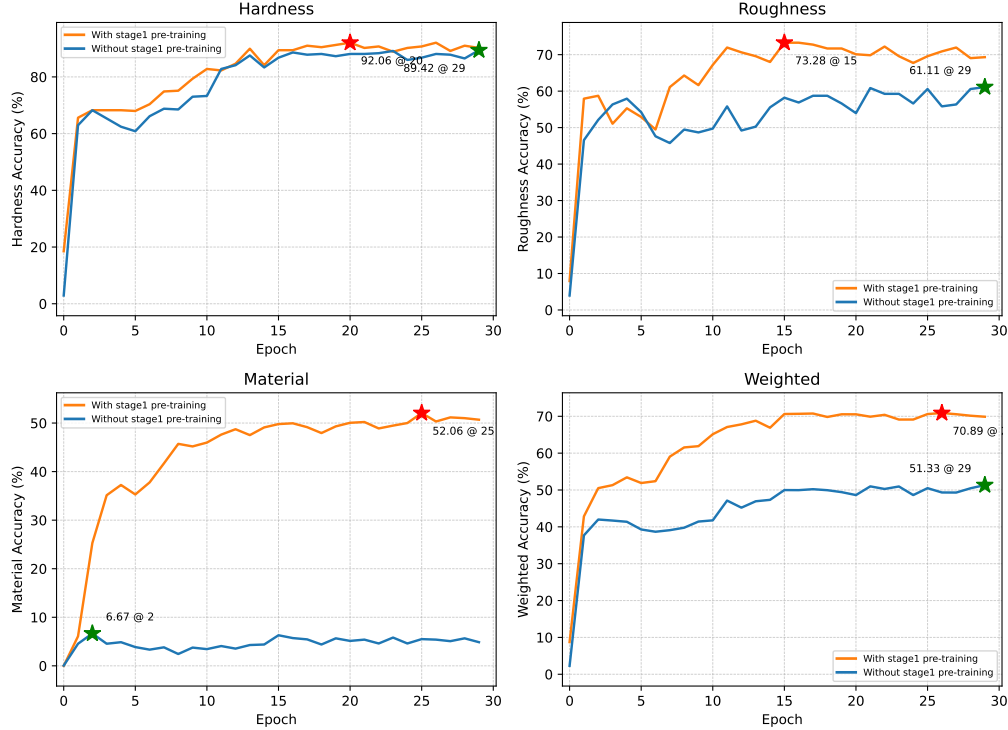


Figure 5: Ablation results on the VitaSet dataset across tasks. Each variant removes one key stage/component from the full model, demonstrating the necessity of explicit alignment and multimodal fusion for robust multi-task property learning.

The ablation results reveal several key findings:

Limitations of unimodal approaches. (1) **Vision-only models** (e.g., Qwen-2.5-VL) tend to rely heavily on appearance cues and pretrained visual priors. However, when physical properties lack stable or distinctive visual signatures—such as material compliance, micro-texture, or hardness—pure visual reasoning becomes unreliable and brittle. (2) **Tactile-only models** (e.g., Octopi) face two key limitations. First, tactile sensors can only capture information from the localized contact region, making it difficult to extract comprehensive object-level features and potentially leading to inaccurate judgments. Second, since these models typically adopt vision-language backbones and are fine-tuned after replacing visual inputs with tactile ones, the absence of explicit cross-modal alignment mechanisms often hinders effective mapping from tactile representations to the language space. This

Table 4: Stage 3 few-shot defect detection with LoRA by varying the number of training samples per object K (6 objects; test set fixed).

Method	Objects	K (train/object)	Accuracy (%)
VitaTouch (Ours, V+T, LoRA)	6	5	45.0
VitaTouch (Ours, V+T, LoRA)	6	10	75.0
VitaTouch (Ours, V+T, LoRA)	6	15	80.0

misalignment prevents the model from leveraging the rich linguistic priors encoded in the pretrained LLM, resulting in degraded performance.

Necessity of Stage 1 cross-modal alignment. Figure 5 shows that removing Stage 1 explicit cross-modal alignment consistently reduces performance across all tasks. This confirms that mapping vision and touch into a shared language-anchored semantic space is critical for enabling effective multi-modal reasoning.

Benefits of full VitaTouch. The full VitaTouch model integrates both global visual context and local tactile evidence through explicit cross-modal alignment, combining both modalities with language-anchored alignment to better leverage pretrained semantic knowledge in the VLM/LLM backbone, yielding more robust property-centric understanding. This demonstrates that multi-modal fusion combined with explicit alignment mechanisms is essential to achieve optimal performance.

5.3.4. Few-shot Defect Detection via LoRA Adaptation

To reflect real manufacturing settings where defect samples are scarce and continuously evolving, we evaluate few-shot defect detection via LoRA adaptation under limited labels. We fix 6 objects and vary the number of labeled training samples per object (K), while keeping the test set fixed across different K . During adaptation, perception encoders and both Q-Formers are frozen, and only LoRA parameters in the LLM are updated, enabling efficient tuning under limited compute and memory budgets. Table 4 shows that accuracy improves substantially as K increases, indicating effective transfer from the learned property foundation to defect discrimination, and that LoRA provides a practical, data-efficient pathway for industrial adaptation without full-model finetuning.

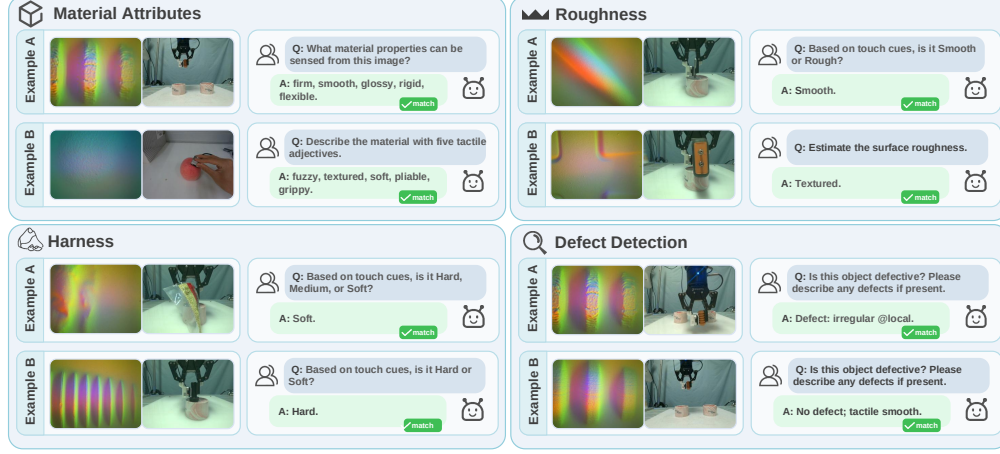


Figure 6: **Qualitative inspection-style outputs of VitaTouch.** We show representative predictions for material attribute description (controlled vocabulary), roughness and hardness classification, and defect decision with brief description.

5.3.5. Qualitative Results: Inspection-style Outputs

Figure 6 illustrates inspection-style qualitative outputs of VitaTouch on four representative tasks, including controlled-vocabulary material attributes, roughness and hardness classification, and defect decision with brief description. The examples show that the model follows the required QA format and produces actionable outputs for operator-facing inspection workflows.

5.3.6. Proof-of-Concept: Robotic Sorting Demonstration

To validate deployment feasibility, we implement a proof-of-concept robotic sorting experiment that emulates industrial quality-inspection workflows (Fig. 7). A robotic arm grasps each object, captures aligned vision and tactile observations during contact, and then invokes our Stage 3 LoRA-adapted model to predict whether a surface defect is present. The predicted binary decision is used to trigger a closed-loop sorting action: defective items are placed into the left bin, while normal items are placed into the right bin. This demonstration evaluates end-to-end integration under real sensing noise, contact variability, and execution disturbances, and highlights that the model outputs are directly actionable for downstream automation. Overall, this proof-of-concept demonstrates that VitaTouch can produce actionable defect decisions in controlled laboratory settings, validating technical feasibility and providing a

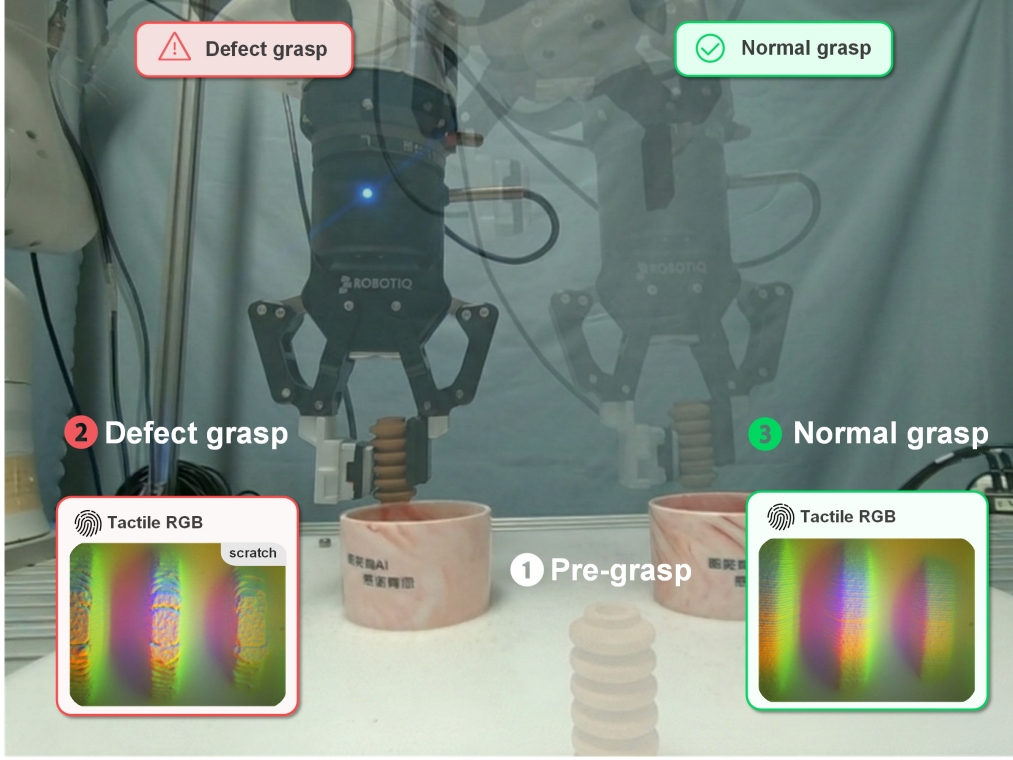


Figure 7: Proof-of-concept robotic sorting demonstration. The robot performs pre-grasp, acquires tactile RGB observations during grasp, predicts defect/normal using the Stage 3 model, and sorts the object into defect (left) or normal (right) bin accordingly.

pathway toward industrial deployment.

6. Conclusion

We presented **VitaTouch**, a property-aware vision–tactile–language foundation model for manufacturing quality inspection. VitaTouch distills visual and tactile evidence via dual Q-Formers and conditions a frozen LLM decoder, enabling natural-language-driven property reasoning and defect inspection beyond appearance-only pipelines. Experiments demonstrate strong effectiveness and transferability: VitaTouch achieves state-of-the-art performance on the TVL benchmark (HCT: 6.10, TVL: 6.09) and delivers high accuracy on VitaSet property tasks (Hardness: 92.06%, Roughness: 73.28%, Material: 52.06%). Moreover, with the proposed progressive three-stage

training paradigm—cross-modal alignment, property foundation learning, and parameter-efficient LoRA adaptation—VitaTouch achieves robust few-shot defect adaptation, reaching 80% accuracy with only 15 labeled samples per object. **To support reproducibility and future research, we will fully release VitaSet on HuggingFace and open-source the training code and model weights upon acceptance.**

Despite these encouraging results and proof-of-concept robotic sorting demonstrations, our evaluation is still limited to 186 objects and controlled laboratory settings. Real shop-floor deployment involves broader product variants, long-tail defect distributions, environmental variability, and long-horizon operational reliability. Future work will focus on scaling data collection with industrial partners, validating across diverse production lines, and addressing system integration challenges with existing manufacturing execution systems.

Overall, VitaTouch establishes property-aware multimodal grounding as a reusable foundation for smart manufacturing inspection, highlighting the potential of integrating tactile sensing with vision-language models for data-efficient and resource-aware industrial automation.

Acknowledgments

This work was supported by the Fundamental Research Funds for the Central Universities (Grant No. 2024PTB-007); the Fundamental Research Funds for Beijing University of Posts and Telecommunications (Grant No. 2025TSQY12); and Zhongguancun Academy (Grant No. 20240307).

Appendix A. Data Collection Prompts

This appendix presents the prompt templates used during VitaSet annotation with GPT-4o API. All prompts follow a structured format with task-specific rules derived from tactile perception principles.

Appendix A.1. Hardness Classification Prompt

You are a tactile perception expert. You receive a vision image, a tactile image, plus the object name and description.

Classify hardness into ONE label (binary):

- Hard → rigid, barely deforms

- Soft → clearly compressible

Key cues (VERY IMPORTANT):

- Primary rule: If the tactile contact boundary/edges look sharp, crisp, and well-defined, the object is likely Hard.
- Primary rule: If the tactile contact boundary/edges look blurred, diffuse, or not clearly defined, the object is likely Soft.
- Use the vision image as a consistency check:
 - * If tactile edges are sharp AND the vision image shows little/no visible deformation, choose Hard (high confidence).
 - * If tactile edges are blurred AND the vision image shows obvious deformation/indentation, choose Soft (high confidence).
 - * If cues conflict (e.g., sharp edges but visible deformation), still follow the tactile-edge rule as the primary signal, but use the object name/description to break ties.

Use visual cues, gel deformation, and text context.

Object: {object_name}

Description: {description}

Reply with exactly one word: Hard or Soft.

Appendix A.2. Roughness Classification Prompt

You analyze surface texture from a vision image, a tactile image, and the object info below.

Choose ONE label:

- Smooth → even, flat surface with little/no bumps
- Textured → regular/repeating raised pattern (structured bumps/ridges)
- Rough → clearly bumpy/irregular micro-relief, but WITHOUT a regular pattern

Key cues (VERY IMPORTANT):

- If the contact surface looks flat and even, choose Smooth.
- If there are small bumps/relief that are obvious but irregular and non-repeating, choose Rough.

- If the raised pattern is regular/repeating (organized texture), choose Textured.

Consider tactile detail, surface micro-structure, and the description.

Object: {object_name}

Description: {description}

Reply with exactly one word: Smooth, Textured, or Rough.

Appendix A.3. Material Property Description Prompt

You are a tactile perception expert. You receive: (1) a vision image, (2) a tactile image, (3) the object name and a short description.

Task: Select ~5 tactile adjectives (prefer EXACTLY 5) that best describe the surface/material properties.

IMPORTANT: You MUST choose ONLY from this fixed vocabulary (case-insensitive), and you MUST NOT invent new words:
hard, firm, soft, cushioned, rigid, flexible, smooth, rough, grainy, fibrous, uneven, patterned, glossy, matte, slippery, sticky, grippy, metallic, plastic, woven

Rules:

- Output 3 to 5 words (prefer 5).
- Use only the words from the list above.
- Do not output temperature, color, size, shape, or object name.
- Comma + space separated. Lowercase. End with a period.
- Choose the closest matches based on BOTH images and the text info.

Object: {object_name}

Description: {description}

Output:

Appendix A.4. Human Verification Protocol

All GPT-4o generated labels underwent complete object-level human verification across all 186 objects. Reviewers corrected inconsistencies, implau-

sible descriptions, and format violations to ensure label quality and inter-annotator agreement.

References

- [1] A. M. Reis, A. Dall-Orsoletta, E. Nunes, L. Costa, and S. Sousa, “Quality costs and Industry 4.0 inspection strategy modelling and analysis: a system dynamics approach,” *The International Journal of Advanced Manufacturing Technology*, 2024. doi: 10.1007/s00170-024-13184-9.
- [2] J. Liu, G. Xie, J. Wang, S. Li, C. Wang, F. Zheng, and Y. Jin, “Deep industrial image anomaly detection: A survey,” *Machine Intelligence Research*, vol. 21, no. 1, pp. 104–135, 2024. doi: 10.1007/s11633-023-1459-z.
- [3] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, “MVTec AD – A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [4] Y. Ma, J. Yin, F. Huang, and Q. Li, “Surface defect inspection of industrial products with object detection deep networks: a systematic review,” *Artificial Intelligence Review*, vol. 57, art. 333, 2024. doi: 10.1007/s10462-024-10956-3.
- [5] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning Transferable Visual Models From Natural Language Supervision,” *arXiv preprint* arXiv:2103.00020, 2021. Available: <https://arxiv.org/abs/2103.00020>
- [6] J. Li, D. Li, S. Savarese, and S. Hoi, “BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models,” in *International Conference on Machine Learning (ICML)*, 2023. Available: <https://arxiv.org/abs/2301.12597>
- [7] W. Dai, J. Li, D. Li, A. M. H. Tiong, J. Zhao, W. Wang, B. Li, P. Fung, and S. Hoi, “InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning,” *arXiv preprint* arXiv:2305.06500, 2023. Available: <https://arxiv.org/abs/2305.06500>

- [8] H. Liu, C. Li, Q. Wu, and Y. J. Lee, “Visual Instruction Tuning,” *arXiv preprint* arXiv:2304.08485, 2023. Available: <https://arxiv.org/abs/2304.08485>
- [9] X. Jiang, J. Li, H. Deng, Y. Liu, B.-B. Gao, Y. Zhou, J. Li, C. Wang, and F. Zheng, “MMAD: The First-Ever Comprehensive Benchmark for Multimodal Large Language Models in Industrial Anomaly Detection,” *arXiv preprint* arXiv:2410.09453, 2024. Available: <https://arxiv.org/abs/2410.09453>
- [10] Y.-H. Xin, K.-M. Hu, R.-J. Xiang, Y.-L. Gao, J.-F. Zhou, G. Meng, and W.-M. Zhang, “Vision-based tactile sensing: From performance parameters to device design,” *Applied Physics Reviews*, vol. 12, no. 2, art. 021312, 2025. doi: 10.1063/5.0249440.
- [11] R. Feng, J. Hu, W. Xia, T. Gao, A. Shen, Y. Sun, B. Fang, and D. Hu, “AnyTouch: Learning Unified Static-Dynamic Representation across Multiple Visuo-tactile Sensors,” *arXiv preprint* arXiv:2502.12191, 2025. doi: 10.48550/arXiv.2502.12191. Available: <https://arxiv.org/abs/2502.12191>
- [12] T. Defard, A. Setkov, A. Loesch, and R. Audigier, “PaDiM: a Patch Distribution Modeling Framework for Anomaly Detection and Localization,” in *Proc. Int. Conf. Pattern Recognition (ICPR)*, 2021.
- [13] V. Zavrtanik, M. Kristan, and M. Skočaj, “DRAEM – A discriminatively trained reconstruction embedding for surface anomaly detection,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021.
- [14] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, “Towards Total Recall in Industrial Anomaly Detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022. Available: <https://arxiv.org/abs/2106.08265>
- [15] Y. Zou, J. Jeong, L. Pemula, D. Zhang, and O. Dabeer, “SPot-the-Difference Self-Supervised Pre-training for Anomaly Detection and Segmentation,” in *Proc. European Conf. Computer Vision (ECCV)*, 2022. Available: <https://arxiv.org/abs/2207.14315>
- [16] C. Wang, W. Zhu, B.-B. Gao, Z. Gan, J. Zhang, Z. Gu, S. Qian, M. Chen, and L. Ma, “Real-IAD: A Real-World Multi-View Dataset

- for Benchmarking Versatile Industrial Anomaly Detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [17] J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, and O. Dabeer, “WinCLIP: Zero-/Few-Shot Anomaly Classification and Segmentation,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2023. Available: <https://arxiv.org/abs/2303.14814>
 - [18] X. Li, Z. Zhang, X. Tan, C. Chen, Y. Qu, Y. Xie, and L. Ma, “PromptAD: Learning Prompts with only Normal Samples for Few-Shot Anomaly Detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024. Available: <https://arxiv.org/abs/2404.05231>
 - [19] Q. Zhou, G. Pang, Y. Tian, S. He, and J. Chen, “AnomalyCLIP: Object-agnostic Prompt Learning for Zero-shot Anomaly Detection,” in *International Conference on Learning Representations (ICLR)*, 2024. Available: <https://arxiv.org/abs/2310.18961>
 - [20] Z. Qu, X. Tao, M. Prasad, F. Shen, Z. Zhang, X. Gong, and G. Ding, “VCP-CLIP: A Visual Context Prompting Model for Zero-shot Anomaly Segmentation,” *arXiv preprint arXiv:2407.12276*, 2024. Available: <https://arxiv.org/abs/2407.12276>
 - [21] Y. Cao, J. Zhang, L. Frittoli, Y. Cheng, W. Shen, and G. Boracchi, “AdaCLIP: Adapting CLIP with Hybrid Learnable Prompts for Zero-Shot Anomaly Detection,” in *Proc. European Conf. Computer Vision (ECCV)*, 2024. Available: <https://arxiv.org/abs/2407.15795>
 - [22] S. Li, J. Cao, P. Ye, Y. Ding, C. Tu, and T. Chen, “Clip-SAM: CLIP and SAM Collaboration for Zero-Shot Anomaly Segmentation,” *arXiv preprint arXiv:2401.12665*, 2024. Available: <https://arxiv.org/abs/2401.12665>
 - [23] W. Ma, X. Zhang, Q. Yao, F. Tang, C. Wu, Y. Li, R. Yan, Z. Jiang, and S. K. Zhou, “AA-CLIP: Enhancing Zero-shot Anomaly Detection via Anomaly-Aware CLIP,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2025. Available: <https://arxiv.org/abs/2503.06661>

- [24] T. Gong, Q. Chu, B. Liu, W. Zhou, and N. Yu, “FE-CLIP: Frequency Enhanced CLIP Model for Zero-Shot Anomaly Detection and Segmentation,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2025.
- [25] Z. Gu, B. Zhu, G. Zhu, Y. Chen, M. Tang, and J. Wang, “AnomalyGPT: Detecting Industrial Anomalies Using Large Vision-Language Models,” in *Proc. AAAI Conf. Artificial Intelligence (AAAI)*, 2024. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/27963>
- [26] A. Costanzino, P. Zama Ramirez, G. Lisanti, and L. Di Stefano, “Multi-modal Industrial Anomaly Detection by Crossmodal Feature Mapping,” *arXiv preprint arXiv:2312.04521*, 2023. Note: CVPR 2024. Available: <https://arxiv.org/abs/2312.04521>
- [27] W. Yuan, S. Dong, and E. H. Adelson, “GelSight: High-Resolution Robot Tactile Sensors for Estimating Geometry and Force,” *Sensors*, vol. 17, no. 12, art. 2762, 2017. doi: 10.3390/s17122762.
- [28] M. Lambeta, P.-W. Chou, S. Tian, B. Yang, B. Maloon, Y. Mostajabdeh, D. Stroud, R. Santos, A. Byagowi, D. Jayaraman, and R. Calandra, “DIGIT: A Novel Design for a Low-Cost Compact High-Resolution Tactile Sensor with Application to In-Hand Manipulation,” *IEEE Robotics and Automation Letters*, 2020. Available: <https://arxiv.org/abs/2005.14679>
- [29] A. Agarwal, A. Ajith, C. Wen, V. Stryzheus, B. Miller, M. Chen, M. K. Johnson, J. L. Susa Rincon, J. Rosca, and W. Yuan, “Robotic Defect Inspection with Visual and Tactile Perception for Large-scale Components,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2023. doi: 10.1109/IROS55552.2023.10341590. Available: <https://arxiv.org/abs/2309.04590>
- [30] L. Fu, G. Datta, H. Huang, W. C.-H. Panitch, J. Drake, J. Ortiz, M. Mukadam, M. Lambeta, R. Calandra, and K. Goldberg, “A Touch, Vision, and Language Dataset for Multimodal Alignment,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2024. Available: <https://arxiv.org/abs/2402.13232>
- [31] S. Yu, K. Lin, A. Xiao, J. Duan, and H. Soh, “Octopi: Object Property Reasoning with Large Tactile-Language Models,” in *Robotics: Science and Systems (RSS)*, 2024. Available: <https://arxiv.org/abs/2405.02794>

- [32] C. Finn, P. Abbeel, and S. Levine, “Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2017. Available: <https://arxiv.org/abs/1703.03400>
- [33] J. Snell, K. Swersky, and R. Zemel, “Prototypical Networks for Few-shot Learning,” in *Proc. Neural Information Processing Systems (NeurIPS)*, 2017. Available: <https://arxiv.org/abs/1703.05175>
- [34] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-Rank Adaptation of Large Language Models,” *arXiv preprint* arXiv:2106.09685, 2021. Available: <https://arxiv.org/abs/2106.09685>
- [35] L. Lamport, *L^AT_EX: A Document Preparation System: User’s Guide and Reference Manual*, 2nd ed., Addison-Wesley, 1994.